LESSON 04 · CHEAT SHEET

Lists & Dictionaries

LIST: A COLUMN

```
staff = ["Pell", "Orrin", "Siv"]
staff[0]            'Pell'    # first (from 0)
staff[-1]           'Siv'     # last
staff[1:3]          ['Orrin', 'Siv']
len(staff)          3
"Orrin" in staff    True
staff.append("Bram") # add at the end
staff.remove("Orrin") # remove by value
staff.pop()         # remove & return last
staff.index("Siv")  # position of
[]                  # empty list
```

NUMBERS IN A LIST

```
s = [31000, 36500, 28000]
sum(s)  max(s)  min(s)  len(s)
sum(s) / len(s)              # average
sorted(s)                    # sorted COPY
sorted(s, reverse=True)[:3]  # top 3
s.sort()                     # in place, returns None
```

BUILD A LIST WITH A LOOP

```
above = []                   # before
for x in salaries:
    if x > 30000:
        above.append(x)      # inside
print(above)                 # after
```

DICTIONARY: A ROW WITH HEADERS

```
emp = {"name": "Pell", "salary": 31000}
emp["name"]             'Pell'
emp["salary"] = 32000   # change
emp["rating"] = 4       # add a field
emp["bonus"]            # KeyError!
emp.get("bonus", 0)     0  # safe lookup
"name" in emp           True
len(emp)                3
del emp["rating"]       # remove a field
{}                      # empty dict
```

Keys: exact and case-sensitive. Values: anything.

LOOP OVER A DICTIONARY

```
for key in emp:              # keys
    print(key, emp[key])
for k, v in emp.items():     # both
    print(k, v)
list(emp.keys())  list(emp.values())
sorted(emp)                  # keys, sorted
```

TABLE = LIST OF DICTS

```
staff = [
    {"name": "Pell", "salary": 31000},
    {"name": "Orrin", "salary": 36500},
]
for person in staff:
    print(person["name"])
f"{person['name']}"        # other quotes inside

top = staff[0]
for person in staff:
    if person["salary"] > top["salary"]:
        top = person
```

COUNTIF / GROUPING

```
count = {}
for d in depts:
    if d in count:
        count[d] += 1
    else:
        count[d] = 1

groups = {}                  # dict of lists
for p in staff:
    key = p["dept"]
    if key not in groups:
        groups[key] = []
    groups[key].append(p["hourly"])
```

EXCEL → PYTHON

a column	a list
a row + headers	a dict
a sheet	list of dicts
`XLOOKUP`	`d[key]` / `d.get(key, x)`
`COUNTIF`	counting loop
`SUM/MAX/AVERAGE`	`sum/max`, `sum/len`
Filter	loop + `append`

GOTCHAS

- `.sort()`, `.append()` return `None`
- `d["clockworks"]` ≠ `d["Clockworks"]` : KeyError
- `f"{p["x"]}"` → use `'x'` inside
- `staff[4]` on 4 items: IndexError (last is 3)
- Python prints lists with single quotes. Normal.